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Effect of the geometric dimensionality of computational domain on the results of CFD-modeling of steam methane reforming

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ABSTRACT

A numerical investigation of the steam methane reforming process using RANS is presented. The effect of the geometric dimensionality of the reformer on the numerical results is investigated by performing CFD simulations with 1D, 2D and 3D computational domains. The commercial software ANSYS Fluent is used for this comparative analysis. The numerical study is performed for the wide range of the residence time and the relative length (radius/length). The comparison study of 1D, 2D and 3D models on steam methane reforming is performed for same CFD-code, initial conditions, catalyst type, approximation scheme, the convergence criteria, the turbulence model, the type of solution initialization. The results show the distributions of the mole fraction of the reformate products, temperature, methane conversion rate and diffusion flux inside a 150 mm length and 10 mm radius of steam methane reformer that is filled by nickel based catalyst. The differences between the results for 1D and 3D geometry become insignificant at the residence time of about 8 $kg_{cat}s/mol_{CH4}$ and the relative length of 15. The difference between the results for 2D and 3D geometry is not significant as for 1D and 3D geometries. Almost similar results are achieved with residence time 4 $kg_{cat}s/mol_{CH4}$ and relative length 15. Therefore, for engineering calculations of steam methane reforming, it is sufficient to use a 1D model if the residence time is more than 8 $kg_{cat}s/mol_{CH4}$ and the relative length is more than 15. The 3D and 2D model should be used if the residence time and the relative length have a small value.

Introduction

Steam reforming of hydrocarbons on solid catalysts is the conventional process in petrochemical industry for the hydrogen and synthesis gas (syngas) production. Among the various hydrocarbons, catalytic steam reforming of methane, which is main gas component of natural gas, is one of the most substantial and economical process for the hydrogen and syngas production in large-scale industry [1–3]. In addition, recently there has been a trend in the use of clean energy, one of the main parts of which is the use of hydrogen as a fuel. Therefore, the production of hydrogen will be constantly increasing in the coming decades. In addition, the steam methane reforming process can be used in thermochemical waste-heat recuperation systems [4–6].

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hydrogen and synthesis gas (syngas) production [7–9]. Among the various hydrocarbons, catalytic steam reforming of methane, which is main gas component of natural gas, is one of the most substantial and economical process for the hydrogen and syngas production in large-scale industry [10–13]. In addition, recently there has been a trend in the use of clean energy, one of the main parts of which is the use of hydrogen as a fuel [14–17].

In accordance with the B1-Hydrogen Scenario of the European Commission [18], enhanced growth of large-scale hydrogen production in the world will begin after 2030 and will be stimulated by a significant reduction in the cost of hydrogen technologies and increased consumption of hydrogen in the transport sector. And according to that scenario, the main way for hydrogen production will be steam reforming of natural gas.

The chemical reaction of steam methane reforming (SMR) is a well-studied process, both theoretically [19–23] and experimentally [24–27]. Various types of catalysts are used to increase the efficiency of this process. The nickel based catalysts are widely used in large-scale hydrogen industry [23,24,28–30]. Moreover, the noble metals such as ruthenium (Ru) [31], rhodium (Rh) [32], palladium (Pd) [33], iridium (Ir) [19] and platinum (Pt) [34] are used as the active element in catalysts. The steam methane reforming process mainly consists of three reversible reactions [24]: the endothermic reforming reactions (1) and (3), and the slightly exothermic water-gas shift reaction (2):

$$CH_4 + H_2O = CO + 3H_2, -\Delta H_{298} = -206.1 kJ/mol; \quad (1)$$

$$CO + H_2O = CO_2 + H_2, -\Delta H_{298} = +41.15 kJ/mol; \quad (2)$$

$$CH_4 + 2H_2O = CO_2 + 4H_2, -\Delta H_{298} = -165.0 kJ/mol. \quad (3)$$

Also, the reactions (1–3) may be accompanied by other chemical reactions, but the contribution of these reactions to the heat and mass balance is negligible for correct implementation of the SMR process [24,25].

In recent years, a numerical study of the SMR has found wide application among engineers and scientists in different countries. The main advantages of the numerical investigation are the low cost of the study in comparison with the experiment, as well as a evident visualization of the results. According to the NASA (National Aeronautics and Space Administration) report [35–37], approximately 85% of the total computing power of giant-powered computer in all over the world is used to solve problems of computational fluid dynamics (CFD). In addition, the same organization presented two reports of Revolutionary Computational Aerosciences [38] and Vision 2030 [39], which show the prospects for using numerical modeling of various physical and chemical processes. Thus, the Vision 2030 report indicates that the estimated computing capacities for solving the problems of computational fluid dynamics will increase by approximately 10–12% per year in the next two decades. In addition, computational fluid dynamics modeling provides versatility to modify design parameters without the expense of research equipment changes, which brings large money economic and economy of time [40–42]. The CFD-modeling results of various combined chemical and physical phenomena are in well agreement with the experimental data for different reformer geometries, types of catalysts and operation conditions [43–45]. Various commercial and open-source software are used for modeling and simulation of steam methane reforming process, for example, ANSYS Fluent [46,47], Comsol Multiphysics [48], Autodesk Simulation CFD [49], OpenFOAM [50] (open source). Some researchers are developing their own code to solve the problem of numerical simulation of steam methane reforming process [45].

Lao et al. [43] presented a computational fluid dynamics (CFD) model of a large-scale SMR process for hydrogen production. A developed CFD-model is implemented in ANSYS Fluent for real computational domain to predict temperatures and mole fractions of each element into reformer. The simulation results can be used to improve the energy efficiency of SMR process, and also can be applied to develop efficient reactors. Two-dimensional axisymmetric reformer geometry was used for solving the model. The study was carried out for various operating parameters: inlet gas velocity, feed composition of the reaction mixture, etc. However, the study [43] does not provide a reasoned justification for choosing the geometric dimensionality of computational domain and there is no estimation of the effect of geometric dimensionality on the result of modeling.

Tran and co-authors [46] reported the results of CFD modeling of the typical transport and chemical reaction phenomena observed in an industrial-scale steam methane reformer. The results of that CFD simulation was correlated with results of different literature sources. In that work, three-dimensional geometry was used for modeling. The use of 3D geometry led to an increase in the processing power. The CFD results were calculated by the ANSYS Fluent parallel solver after 3 days on the 80 private cores on UCLAs Hoffman2 Cluster.

Hoang and co-workers [45] suggested simplifying of the CFD model of steam methane reforming process by assuming that the gas flow in the reformer is uniform and apply one-dimensional geometry of the reformer. However, the radial dispersion of the gas flow due to catalyst particle size of 1.75 mm and the radial heat transfer by heat supply through the reformer wall, may have some effects on overall reforming performance. Authors concluded, to take these effects into account for the numerical model, a two-dimensional computational domain is more appropriate to describe the reforming behavior. The numerical results were determined by the author's code.

As a rule, research studies on CFD simulations of steam methane reforming process are mainly focused on investigation of heat and mass transfer characteristics only for one or two computational domains, for example, either for two-dimensional or for three-dimensional geometry. However, the results of the numerical study are influenced by the nature of the calculated computational geometry [44]. The discrepancy between the calculation results for 1D, 2D and 3D geometry can be significant, especially with a small residence time. In addition, the use of different codes (software) for modeling the SMR process for the same initial conditions and catalyst type can give different results. Also, the results of numerical investigation are significantly influenced by the

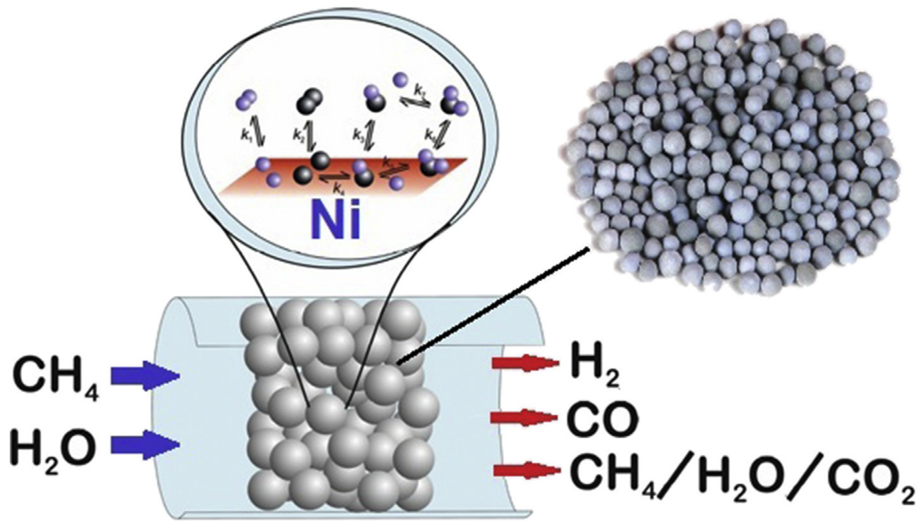

Fig. 1 – The schematic diagram of steam methane reforming process over a Ni-based catalyst.

approximation scheme, the convergence criteria, the turbulence model, the type of solution initialization, and so on [51,52].

There are no literature sources (according to the author's literature review) on comparison study of 1D, 2D and 3D numerical investigations on steam methane reforming characteristics for same CFD-code, initial conditions, catalyst type, approximation scheme, the convergence criteria, the turbulence model, the type of solution initialization. Carrying out such a comparative analysis for different initial conditions will allow researchers and specialists to choose the geometric dimensionality of computational domain that will allow to obtain accurate results while computing power is minimal.

This paper is devoted to comparison of results of CFD investigation of steam methane reforming process for 1D, 2D and 3D computational domain. The investigation was carried out via the commercial software ANSYS Fluent. Effect of the geometric dimensionality of computational domain on the results of CFD-modeling of steam methane reforming is determined for the various initial conditions.

With modern computational tools and facilities, numerical calculations with large codes aimed to predict the performance of devices for steam methane reforming are feasible. Whether they will partly or fully replace experimental investigations will largely depend on the reliability of the chemical and turbulence models used and the comprehensive study of various aspects of numerical simulation of the steam methane reforming process. Therefore, the investigation of the effect of the geometric dimensionality of computational domain on characteristics of steam methane reforming process is a crucial task.

Numerical setup

Computational domain and mesh

Steam methane reforming is studied in a fixed bed reactor. The steam methane reactor is filled with a nickel-based catalyst as depicted in Fig. 1. The shape of the catalyst particles is a filled sphere. For this numerical study, it is assumed

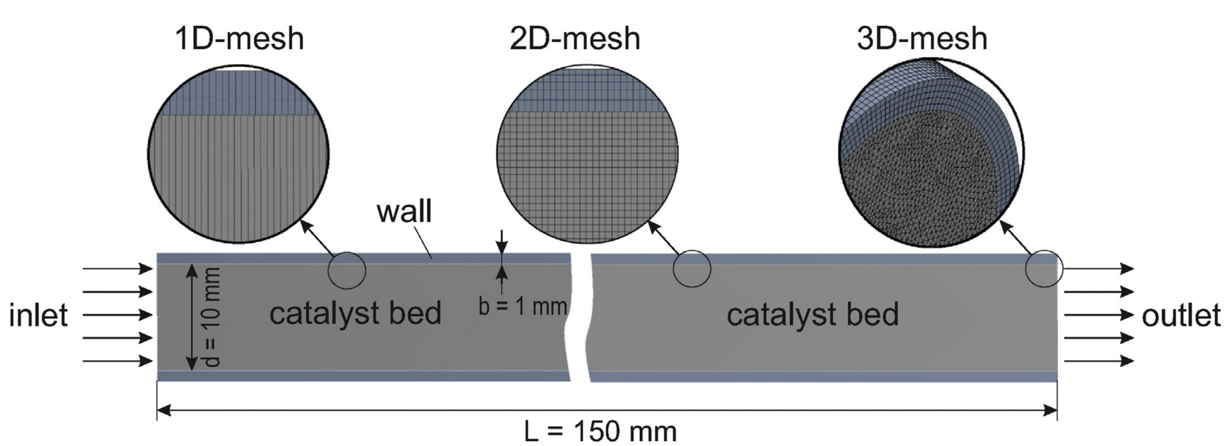

Fig. 2 – The computational domain and mesh for 1D, 2D and 2D geometries.

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Table 1 – Catalyst properties [45].

Catalyst properties	Value
Nickel content (wt%)	9.8
S content (wt%)	4.9
Alumina content (wt%)	85.3
Surface area (m^2/g)	155
Size of the sphere (mm)	1.75
Density (kg/m^3)	1872
Specific heat (J/kg·K)	910
Thermal conductivity (W/m·K)	12.75
Molecular weight (kg/kgmol)	184.41

that the size of all catalyst particles is the same. In reality, the diameter of the sphere can differ by 0.1–0.2 mm. More detailed information about the catalyst and its structure can be obtained in the work of Hoang and co-workers [45].

Chemical reactions occur on the catalyst in the presence of an external supply of heat to the walls of the reaction space of the reformer. The input heat is consumed during endothermic reactions 1 and 3; the residual heat is supplied on heating the mixture. The external heat flow is distributed evenly along the length of the reaction zone.

The computational domain of reformer is presented in Fig. 2. The reformer contains reaction space with catalysts and steel wall. The length of reaction space is 150 mm, the diameter is 10 mm. The thickness of the wall is 1 mm. At the first stage of present investigation for model verification, all geometric characteristics of reformer and catalyst accurately repeat the experiment [45]. The physical property catalyst is shown in Table 1.

The mesh structures for 1D, 2D and 3D geometries presented in Fig. 2. The edge size of elementary cells is the same for all geometries. For 1D mesh the size of cells in reaction space is equal to diameter of reaction space and for wall it is equal to wall thickness. To determine minimum number of elementary cells needed for an accurate and least time consuming simulation, different mesh structures with the different cell densities are constructed. The results of this determination are given below. An additional space of reformer without a catalyst, heat input and with an adiabatic wall is used to form a developed flow at the inlet of the reaction space. This additional space is not shown in Fig. 2.

The catalyst structure of 1D, 2D and 3D geometries is set as porous medium area (for 1D and 2D) and volume (for 3D). The spherical particles of catalyst are taken into account in the steam methane reforming reactor. The Ergun equation is used to derive porous media inputs for a packed bed. The use of this method makes it possible to obtain the same properties of the catalyst layer for different geometries (1D, 2D and 3D). For k-ε model, packed beds are modeled using both a permeability and an inertial loss coefficient. One technique for deriving the appropriate constants involves the use of the Ergun equation [53], a semi-empirical correlation applicable over a wide range of Reynolds numbers and for many types of packing:

$$\frac{|\Delta p|}{L} = \frac{150\mu}{D_p^2}\frac{(1-\varepsilon)^2}{\varepsilon^3}\nu + \frac{1.75\rho}{D_p}\frac{(1-\varepsilon)}{\varepsilon^3}\nu^2 \tag{4}$$

where L – the bed length; D_p – the particle diameter; ε – the total pore volume of particle.

For all directions (Direction-1, Direction-2, Direction-3) viscous resistance coefficient is similar. The viscous resistance coefficient $(1/\alpha)$ for Fluent solver can be defined as:

$$\alpha = \frac{D_p^2}{150}\frac{\varepsilon^3}{(1-\varepsilon)^2} \tag{5}$$

This method for determining the catalytic bed uses the fact that SMR reaction occur on the surface of the solid catalyst particles. In addition, such a method for determining the catalytic bed makes it possible to simplify the heat transfer in this work. The temperature gradient between catalyst adjacent to the wall of the reactor and catalyst at the center of the reactor presented as the temperature gradient between wall and porous medium.

Governing equations

In developed steam methane reforming model the conservation equation, momentum and energy equation in combination with species equations are solved in ANSYS Fluent. The assumptions are taken for CFD-modeling: steady-state conditions of SMR process; no the flux of energy due to a mass concentration gradient (no Duflor effects) [54]; work by viscous forces and by pressure is not done; surface oxidation reactions of the metal wall are absent.

The equation for conservation of mass, or continuity equation, for steady-state conditions can be written as follows:

$$\nabla\cdot(\rho\vec{v}) = 0 \tag{6}$$

For 1D geometry, the continuity equation can be written as follows:

$$\frac{\partial}{\partial x}(\rho v_x) = 0 \tag{7}$$

For 2D geometry the continuity equation has the following form:

$$\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) = 0 \tag{8}$$

And for 3D geometry:

$$\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0 \tag{9}$$

In these equations v_x, v_y and v_z are the axial and radial velocity, respectively; x, y and z are the axial and radial coordinate, respectively.

Conservation of momentum in an inertial (non-accelerating) reference frame is described by follow equations [55]:

$$\nabla\cdot(\rho\vec{v}\vec{v}) = -\nabla\cdot p + \nabla\cdot(\bar{\bar{\tau}}) + \rho\vec{g} + \vec{F} \tag{10}$$

where p is the static pressure; $\rho\vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. The stress tensor $\bar{\bar{\tau}}$ is given by follows equation:

$$\bar{\bar{\tau}} = \mu\left[\left(\nabla\vec{v} + \vec{v}^{\mathrm{T}}\right) - \frac{2}{3}\nabla\cdot\vec{v}I\right] \tag{11}$$

where μ is the molecular viscosity, I is the unit tensor, and the

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second term on the right hand side is the effect of volume dilation.

The conservation equation of energy for reaction space of reformer can be written as follows:

$$\nabla\cdot\vec{u}(p+\rho\cdot h_f)=\nabla\cdot\left[k_{eff}\cdot\nabla T-\left(\sum h_j\cdot\vec{J}_j\right)+\bar{\bar{\tau}}\cdot\vec{v}\right]+S_f^h \tag{12}$$

In this equation T is the reaction mixture temperature; ρ is the density; h_f is enthalpy of gas mixture; $\vec{J}_j$ is the diffusion flux of j element; k_{eff} is the conductivity; h_j is the enthalpy of j element; S_f^h is the fluid enthalpy source term (include the source of energy due to chemical reaction).

For 1D geometry, the energy conservation equation can be written as follows:

$$\frac{\partial(\rho v_x h)}{\partial x}=\frac{\partial}{\partial x}k_{eff}\frac{\partial T}{\partial x}-\frac{\partial\sum h_j J_j}{\partial x}+\bar{\bar{\tau}}\frac{\partial v_x}{\partial x}+S_f^h \tag{13}$$

Conservation equation of energy for steel wall can be written as follows:

$$\nabla\cdot(k_w\cdot\nabla T)=q_w \tag{14}$$

where k_w is the thermal conductivity of the steel wall; q_w is heat through the wall.

Transport of j element of reaction mixture is described by follow equations:

$$\nabla\cdot(\rho\cdot\vec{v}\cdot Y_j)=-\nabla\vec{J}_j+R_j \tag{15}$$

where R_j is chemical reaction rate of species j formation or decomposition; Y_j is the mass fraction of j element of reaction mixture; $\vec{J}_j$ is diffusion flux of j species.

Diffusion flux of j species can be written as:

$$\vec{J}_j=-\left(\rho\cdot D_{j,m}+\frac{\mu_t}{Sc_t}\nabla Y_j-D_{T,j}\frac{\nabla T}{T}\right) \tag{16}$$

where $D_{j,m}$ and $D_{T,j}$ are the mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

Turbulence model

Typically, the steam methane reforming is carried out as the laminar flow of the reaction mixture. The flow around the catalyst particle may have a vortex after particle; thereby using a laminar model may give an error in the results [56]. The choice of turbulence model depends on considerations such as the physics encompassed in the flow, the established practice for a specific class of problem, the level of accuracy required, the available computational resources, and the amount of time available for the simulation. For present numerical model, the RNG k-ε model is chosen [55,57–60].

$$\frac{\partial}{\partial x_i}(\rho k v_i)=\frac{\partial}{\partial x_j}\left(\alpha_k\mu_{eff}\frac{\partial k}{\partial x_j}\right)+G_k+G_b-\rho\varepsilon-Y_M+S_k \tag{17}$$

and

$$\frac{\partial}{\partial x_i}(\rho\varepsilon v_i)=\frac{\partial}{\partial x_j}\left(\alpha_\varepsilon\mu_{eff}\frac{\partial\varepsilon}{\partial x_j}\right)+C_{1\varepsilon}\frac{\varepsilon}{k}(G_k+C_{3\varepsilon}G_b)-C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}-R_\varepsilon+S_\varepsilon \tag{18}$$

In equations for k-ε turbulence model, G_k shows the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. The quantities α_k and α_ε are the inverse effective Prandtl numbers for k and ε, respectively. S_k and S_ε are user-defined source terms.

The additional term R_ε in the ε equation can be determined as follows:

$$R_\varepsilon=\frac{C_\mu\rho\eta^3(1-\eta/\eta_0)}{1+\beta\eta^3}\frac{\varepsilon^2}{k} \tag{19}$$

where $\eta\equiv Sk/\varepsilon$, $\eta_0=4.38$, $\beta=0.012$.

The model constants $C_{1\varepsilon}$ and $C_{2\varepsilon}$ in k-ε turbulence model used by default setup for Fluent solver of ANSYS [61]:

$$C_{1\varepsilon}=1.42; C_{2\varepsilon}=1.68 \tag{20}$$

Kinetic model

The calculations of reaction rate for SMR process in present investigation are based on the nuanced analysis and investigations of chemical reactions that are presented in scientific publication [24,45,62]. It has been argued that both CH_4 and H_2O were adsorbed on the catalyst with dissociation. The expressions for determination of reaction rate suggested by Froment and Xu [24] are employed in this model.

The reaction rate of the methane consumption through reactions (1 and 3) can be found from following expression:

$$r_{CH_4}=R_1+R_3. \tag{21}$$

The reaction rate of carbon monoxide (CO) and carbon dioxide (CO_2) production can be found from following expressions:

$$r_{CO}=R_1-R_2. \tag{22}$$

$$r_{CO_2}=R_3+R_2. \tag{23}$$

The rate of each reaction can be determined from following expressions [45]:

$$R_1=\frac{k_1}{p_{H_2}^{2.5}}\left(p_{CH_4}\cdot p_{H_2O}-\frac{p_{H_2}^3\cdot p_{CO}}{K_{e1}}\right)\cdot\frac{1}{Q_r^2}; \tag{24}$$

$$R_2=\frac{k_2}{p_{H_2}}\left(p_{CO}\cdot p_{H_2O}-\frac{p_{H_2}\cdot p_{CO_2}}{K_{e2}}\right)\cdot\frac{1}{Q_r^2}; \tag{25}$$

$$R_3=\frac{k_3}{p_{H_2}^{3.5}}\left(p_{CH_4}\cdot p_{H_2O}^2-\frac{p_{H_2}^4\cdot p_{CO_2}}{K_{e3}}\right)\cdot\frac{1}{Q_r^2}; \tag{26}$$

$$Q_r=1+K_{CO}\cdot p_{CO}+K_{H_2}\cdot p_{H_2}+K_{CH_4}\cdot p_{CH_4}+\frac{K_{H_2O}\cdot p_{H_2O}}{p_{H_2}}, \tag{27}$$

where p_i – partial pressure of i reacting element, bar; K_{je} – equilibrium constant of 1, 2, 3 reactions, respectively; k_j – kinetic rate constant of j reaction.

In Eq. (27) adsorption constants of i element (K_i) can be determined as:

$$K_i=K_{0i}\cdot e^{-\Delta H_i/(R_0T)}, \tag{28}$$

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where H_i – adsorption enthalpy of i element, J/mol; K_{0i} – constant of i component.

Boundary conditions

The specific heat and density of the reaction mixture are determined by the equations of mixing-law and incompressible-ideal-gas law, respectively. It can be possible, because the temperature of reaction mixture is moderate high (less than 1000 K) and the pressure in the reaction space is moderate low (less than 10 bar). Also Mach number is about 0.001 due to low velocity of reaction mixture. Therefore, the assumption for ideal incompressible gas flow is sufficiently provided. The viscosity of the gas flow is determined by Sutherland viscosity law. The thermal conductivity of the gas flow is computed as average mass fraction of each element. The pressure-based solver is used. The convergence criterion for momentum, continuity and species are defined as 10^{-3}, for energy less than 10^{-6}. The hybrid initialization is chosen to initialize calculation. For described equations the Second-Order Upwind approximation method is used to obtain a higher calculation accuracy. When second-order accuracy is desired, quantities at cell faces are computed using a multi-dimensional linear reconstruction approach [63]. The initial conditions for the model are presented in Table 2.

Moreover, the boundary conditions for the computational domain of the reformer shown in Fig. 2 are determined as follows:

- reformer inlet:

$$x = 0; T_g = T_{g(in)}; C_i = C_{i(in)}. \tag{29}$$

- reformer outlet:

$$x = L; \frac{\partial C_i}{\partial x} = 0. \tag{30}$$

- reformer center:

$$y = 0; \frac{\partial C_i}{\partial y} = 0; \frac{\partial T_g}{\partial y} = 0. \tag{31}$$

where C_i is mole concentration of i element.

Results and discussion

Grid independence study

To determine minimum number of cells needed for an accurate and least time consuming simulation, different mesh structures with different mesh densities are constructed. Steam methane reformer model, which included these different mesh structures, are modeled under the same boundary conditions. Hydrogen mole fraction profiles are compared. In Fig. 3 predicted centerline profiles of hydrogen mole fraction for various meshes can be seen.

In accordance with the goals of present investigation, the number of mesh elements and its size cells for 2D geometry was determined, in which a further increase in their number would not have a significant effect on the final results. Based on the profiles of hydrogen mole fraction, the following facts can be concluded: for two-dimensional geometry, increasing the number of mesh cells above 100,000 does not significantly effect on the final results. The size of edge of elementary cell is $2 \cdot 10^{-2}$ mm. Therefore, to save computational time and use same mesh structure for all geometries the mesh structures with 1230 cells for 1D computational domain, 102,378 cells for 2D computational domain and 2,340,761 cells for 3D computational domain were chosen.

Table 2 – Initial conditions.

No.	Parameter	Inlet	Outlet
1	Steam-to-methane ratio	var	–
2	Temperature	var	–
3	Turbulent intensity	5%	5%
4	Hydraulic diameter	10^{-2} m	10^{-2} m
6	Gauge pressure	1.5 bar	0.0 bar

Model validation

To validate and verify developed CFD model, numerical results were compared with experimental investigation of Houng et al. [45] for various initial conditions. The initial conditions for SMR process in present investigation study are set as follow: the temperature in the reaction space is 873 and 1073 K, the reformer pressure is regulated at 1.5 bars; residence time is about 3.6 $kg_{cat}s/mol_{CH4}$ of inlet methane and the steam-to-methane ratio is from 2 to 5. The mass of catalyst loaded in the reformer is 8.98 g, the diameter of catalyst particle is 1.75 mm. Comparisons of experimental data and numerical results of hydrogen yield at residence time of 3.59 $kg_{cat}s/mol_{CH4}$ is shown in Fig. 4.

The discrepancies between numerical and experimental results can be seen for all investigated geometries. The maximum discrepancies of the results are observed for a one-dimensional model. The minimum discrepancies in the results are observed for 2D geometry at 873 K and for 3D geometry at 1073 K. The maximum deviation at 873 K are 11.2% for 1D geometry, 4.6% for 2D geometry and 4.8% for 3D

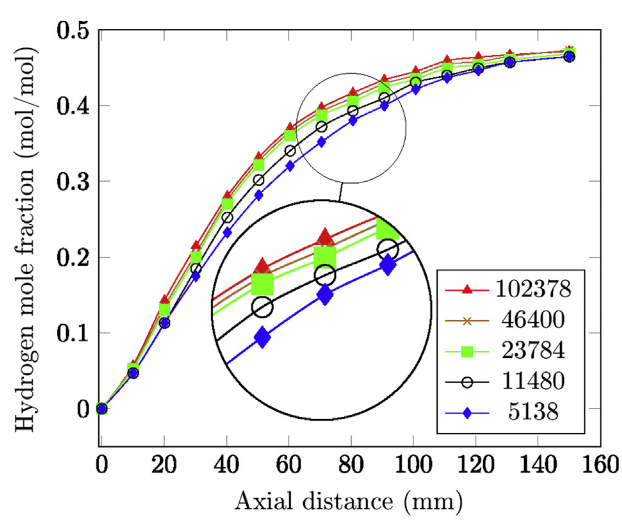

Fig. 3 – Grid independence study for 2D computational domain.

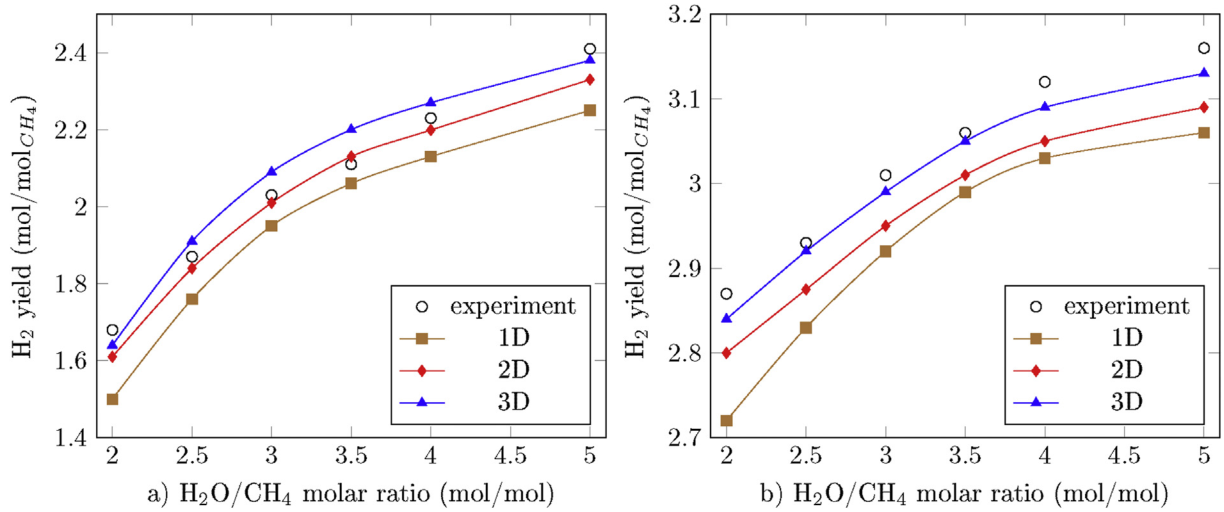

Fig. 4 – Model validation: comparison of experimental data and numerical results: a) T = 873 K; b) T = 1073 K.

geometry; at 1073 K are 7.3% for 1D, 4.1% for 2D and 2.1% for 3D. Such discrepancies can be explained by the fact that at a lower temperature the reaction rate of steam methane reforming is less than at a higher temperature. Therefore, at 873 K, the discrepancy between the results for different geometries is less than at 1073 K.

Moreover, the numerical results of the carbon monoxide mole fraction at the reformer outlet were compared with experimental data [45]. In Fig. 5 it can be seen good correlation between experimental and numerical results.

Based on these results, it can be concluded that developed CFD model for three types of computational domain of steam methane reformer is reliable to investigate the effects of geometric dimensionality of computational domain on the results of CFD-modeling of steam methane reforming process.

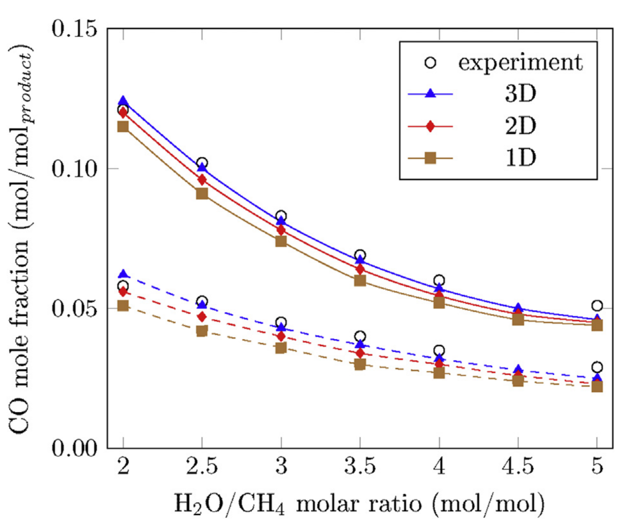

Fig. 5 – Model validation: comparison of experimental data and numerical results: dash line is at T = 873 K; solid line is at T = 1073 K.

H_2 and CO mole fraction

Using a parallel computational environment with message passing interface technology, the simulation of 1D CFD model is converged in about 2 min, 2D model is converged in about 10 min, and 3D model in about 3 h. The CFD models were solved in ANSYS Fluent 16.5 (Academic Research version) on Xeon E5-2670 Sandy Bridge-EP server with single 8-core 2.6 GHz processors with 32 GB RAM. The steady-state results of the CFD modeling are depicted in Fig. 6. Due to the small diameter-to-length ratio of the computational domain, in all contours of the numerical results, the radius is scaled up by 6 times which is convenient and is done for display purposes only.

Fig. 6 shows the distribution of hydrogen mole fraction for 1D, 2D and 3D computational domains of steam methane reformer. The absence of a radial change of the hydrogen mole fraction occurs for one-dimensional geometry. The contour of the hydrogen mole fraction for the three-dimensional geometry is given for the XY-plane. The contour of hydrogen mole fraction was obtained for steam-to-methane ratio of 2:1. Along the length of the reaction space the mole fraction of H_2 increases as well as the mole fraction of CO, while CH_4 and H_2O mole fraction decrease. As can be seen from Fig. 6, the differences between H_2 mole fraction close to the hot reformer wall and those in the central core of the bed are observed for 2D and 3D geometries. The reasons of such differences of radial gradients caused by the effect of the radial temperature on the reaction rates and the flow adhesion to the hot wall (the reaction mixture tends to zero near the wall). Moreover, the radial differences of H_2 mole fraction are caused by the temperature gradients.

To better understand discrepancy between H_2 mole fraction contours (Fig. 6), the hydrogen mole fraction profile at centerline of the reaction space (left side) and discrepancy between results (right side) are depicted in Fig. 7. The initial conditions for these results are set as following: T = 973 K and residence time is 3.6 $kg_{cat}s/mol_{CH4}$, H_2O:CH_4 = 2:1. Fig. 7 shows that maximum difference between results, which are

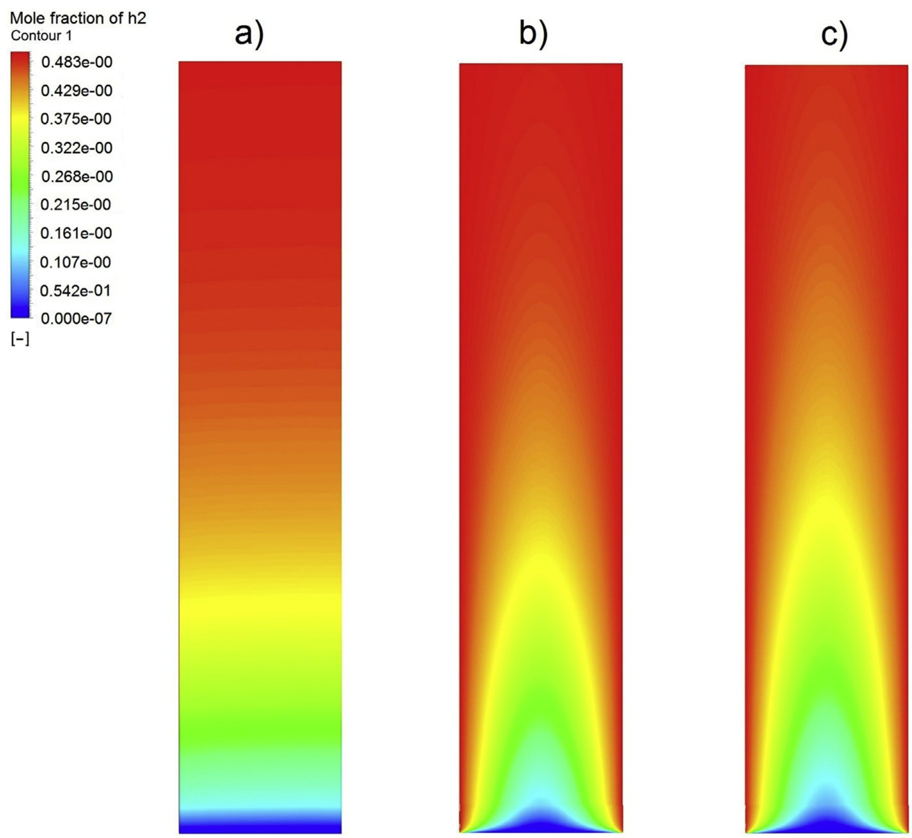

Fig. 6 – Hydrogen mole fraction contour from the reformer CFD simulation at T = 973 K and residence time is 3.6 $kg_{cat}s/mol_{CH4}$.

obtained for 1D, 2D and 3D computational domain, is observed near reformer inlet. The minimum difference between H_2 mole fraction is observed near reformer outlet. Obviously, as the length of the reaction space of the reformer increases, the difference between the outlet results decreases. To better understand discrepancy between the profiles of hydrogen mole fraction, the difference of values is presented in right side of Fig. 7.

Gas flow temperature

An external supply of heat to the reaction space is required to carry out the chemical reaction of the steam methane reforming, which is an endothermic reaction. In numerical simulation, it was assumed that the heat flux is fed to the reaction space through the wall uniformly along the length of the reformer with the heat-flux density $q_w = 2.7\ W/m^2$. Such a

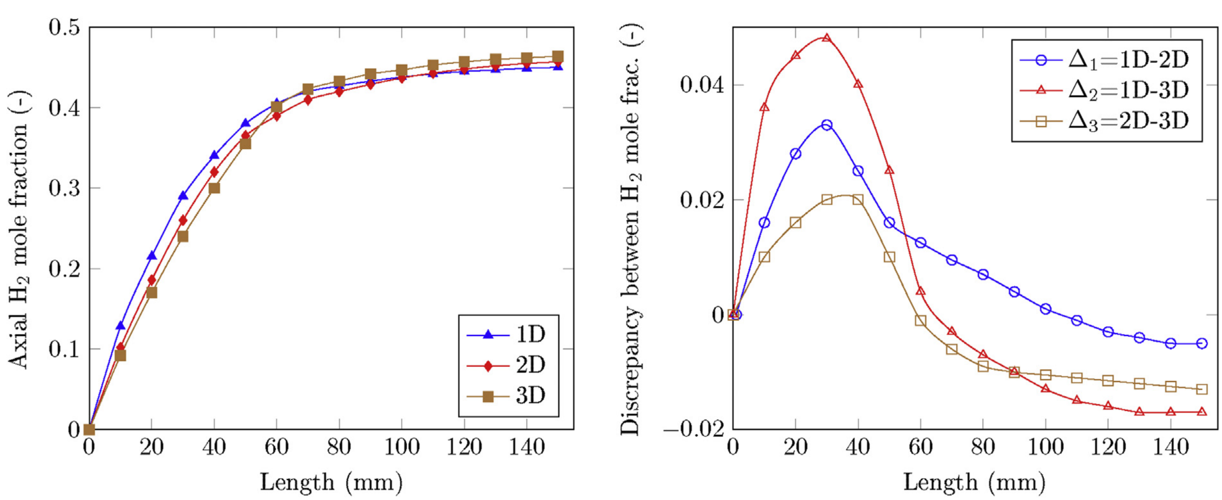

Fig. 7 – Hydrogen mole fraction profile at axial line of the reformer (left side) and discrepancy between results (right side).

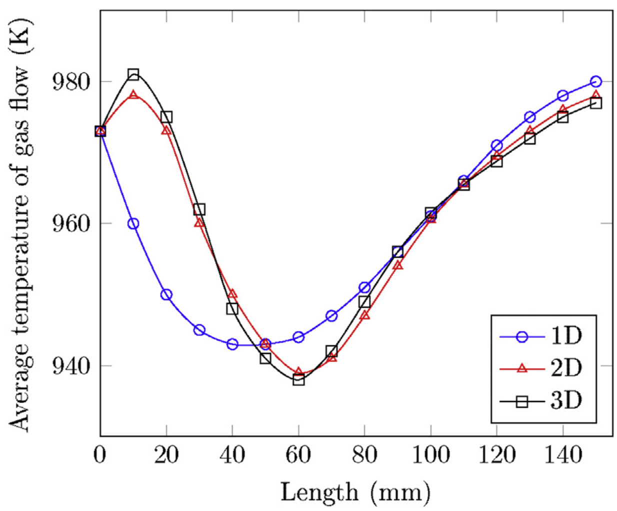

Fig. 8 – Average temperature of gas flow in reformer at T = 973 K, res. time 3.59 $kg_{cat}s/mol_{CH4}$, $H_2O:CH_4 = 2:1$.

heat flux density is sufficient to ensure that the temperature of the reaction mixture at the reformer outlet varies insignificantly compared to the temperature of reaction mixture at the reformer inlet.

Fig. 8 shows the distribution of the average temperature of gas flow along length of the reformer. At the initial time, the temperature of the gas flow is equal to the catalyst temperature. Other initial conditions for these results are set as follow: residence time is 3.59 $kg_{cat}s/mol_{CH4}$, $H_2O:CH_4 = 2:1$.

According to Fig. 8, the average temperature of the gas flow varies along the length of the reformer nonmonotonically for all geometries. For 2D and 3D geometries the profiles of temperature are almost similar. In the initial section near reformer inlet, the temperature increases, since the reaction rates of reactions 1 and 3 are small and the value of heat absorbed during these reactions is less than the value of heat delivered through the wall. Then there is an area in which the temperature decreases due to the high rate of endothermic reactions 1 and 3. At the reformer outlet, the temperature of the mixture increases, which is due to a decrease in reaction rates of reactions 1 and 3 with a decrease in methane mole fraction and an increase in steam-to-methane ratio in the area near to the reformer inlet. The temperature of outer wall is set in Fluent solver as 1000 K for all geometries. To simplify Fig. 8, the constant wall temperature is not shown.

The average temperature profile for the one-dimensional geometry at the initial area near the reformer inlet differs significantly from the temperature profiles for 2D and 3D geometries. There is a sharp drop in temperature in the area near the reformer inlet. On the other hand, in the area near the reformer outlet, the temperature profiles for all geometries practically coincide. The difference in the profiles at the initial area can be explained by the high rate of reactions 1 and 3 that gives a radial temperature gradient. The radial temperature gradient does not take into account the heat balance equation for the one-dimensional model. The main reason of radial temperature gradient is diffusion of *j* element of reaction mixture.

To better understand the effect of diffusion of *j* element on the profiles of average gas flow temperature, Fig. 9 is presented. Fig. 9 shows the deviations between the average temperatures for 2D and 1D, and for 3D and 1D geometries for various residence time: 3.59 $kg_{cat}s/mol_{CH4}$ (solid line), at 10 $kg_{cat}s/mol_{CH4}$ (dash line). For presented computational domain of reformer in Fig. 1, the average velocity at inlet is 0.71 m/s (at 3.59 $kg_{cat}s/mol_{CH4}$) and 0.255 m/s (at 10 $kg_{cat}s/mol_{CH4}$). According to Eq. (15), as the gas flow velocity increases, the influence of diffusion on heat and mass transfer processes decreases. This fact corresponds closely with Fig. 9.

Diffusion flux of methane

The discrepancy between results for 1D, 2D and 3D geometries shown in all figures above can be explained by the mass diffusion of *j* elements and thermal diffusion. Mixing and more generally diffusion phenomena, that takes place at

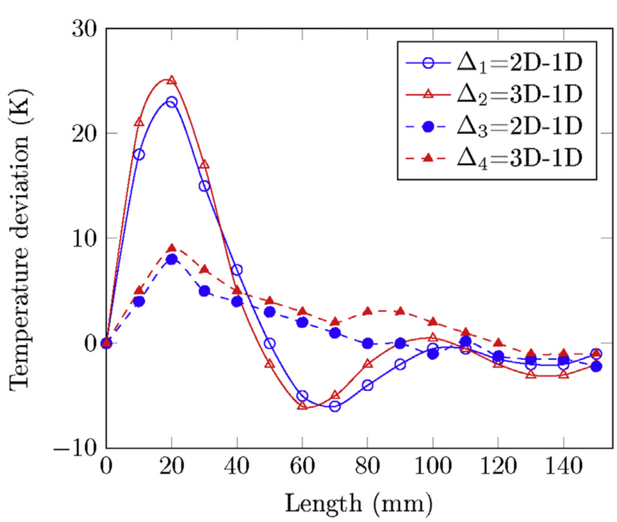

Fig. 9 – Deviation of temperature of gas flow in reformer at T = 973 K, $H_2O:CH_4 = 2:1$ and at 3.59 $kg_{cat}s/mol_{CH4}$ (solid line), at 10 $kg_{cat}s/mol_{CH4}$ (dash line.).

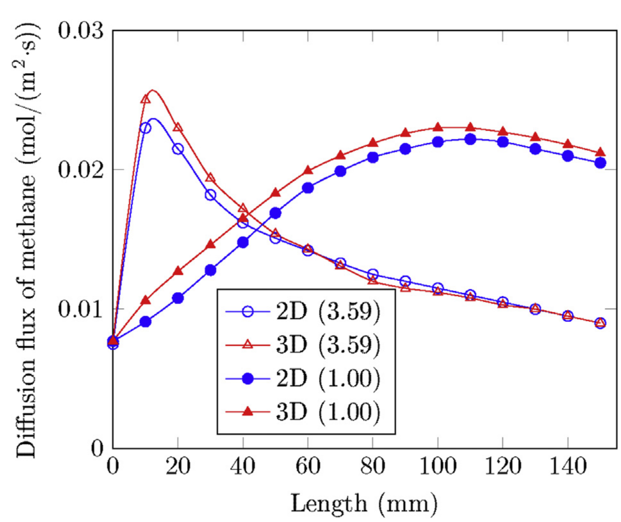

Fig. 10 – Diffusion flux of methane in reformer at T = 973 K, $H_2O:CH_4 = 2:1$ and at 3.59 $kg_{cat}s/mol_{CH4}$ (hollow point), at 1.00 $kg_{cat}s/mol_{CH4}$ (painted point).

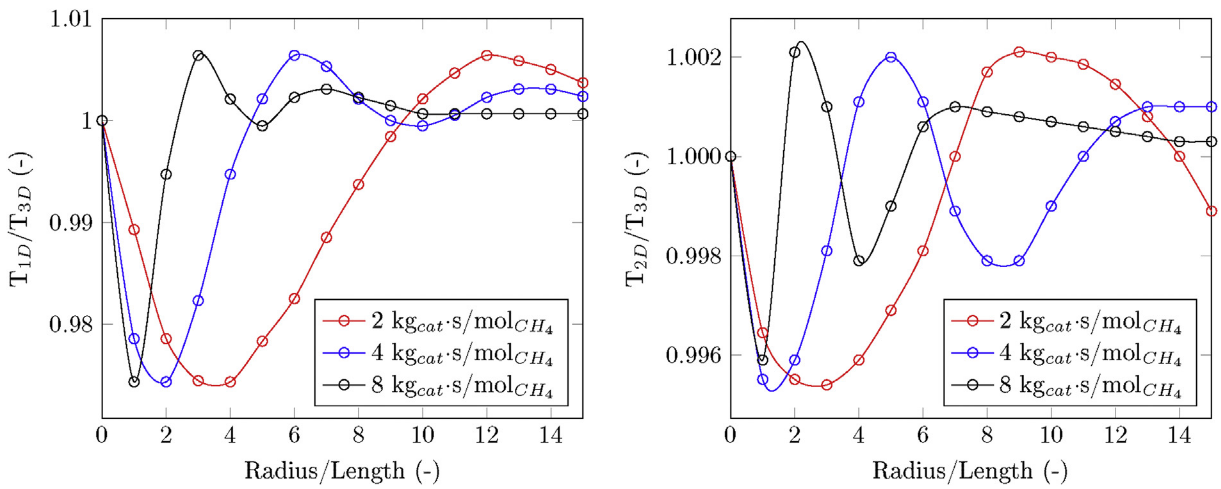

Fig. 11 – Gas flow temperature versus residence time at T = 973 K, H_2O:CH_4 = 2:1.

steam methane reforming process, are never purely 2D and even less 1D. 3D diffusion is accounted by the transport equation of turbulence quantities for the used RANS model and for laminar conditions every parameters depends on geometry of computational domain. To understand effect of residence time and velocity of the reaction gas mixture on the diffusion flux of methane, Fig. 10 is presented.

In Fig. 10 the distribution of the diffusion flux density of methane along the normal to the wall is presented. At residence time 3.59 $kg_{cat}s/mol_{CH4}$, the density of the diffusion flux of methane at the area near the reformer inlet increases sharply, and then, reaching a maximum, decreases smoothly. At residence time 1.00 $kg_{cat}s/mol_{CH4}$, the density of the diffusion flux increases monotonically and reaches its maximum value near the output of the reformer. The dependencies in Fig. 10 show that the diffusion of methane has a significant effect on the results of calculations especially at low residence time.

Effect of geometric dimensionality

Fig. 11 shows the ratio of the average gas flow temperature for the one-dimensional model to the average temperature for the three-dimensional model (left side) and the ratio of the average gas-flow temperature for the two-dimensional model to the average temperature for the three-dimensional model (right side). From Fig. 11 it can be seen that the difference in temperature for the 1D and 3D models is influenced both by the relative length of the reformer and by the residence time.

In the area near the reformer inlet, there is a significant difference in temperatures. As the relative length increases, these discrepancies decrease. Also, as residence time increases, these discrepancies also decrease. It can be concluded that for a residence time more than 8 $kg_{cat}s/mol_{CH4}$ and a relative length of more than 15, it is sufficient to use a simpler one-dimensional model.

The temperature differences for the two-dimensional and three-dimensional models are much smaller than for the one-dimensional and three-dimensional models. However, as it can be seen from Fig. 11, for low residence time in the area near the reformer inlet, there is a significant difference between the temperatures. In general, it can be argued that for engineering calculations if the residence time is greater than 3–4 $kg_{cat}s/mol_{CH4}$ and the relative length is greater than 15, it is sufficient to use a two-dimensional model instead of a three-dimensional one.

Conclusion

A numerical study on the steam methane reforming process is conducted. The SMR process was investigated by CFD-modeling for different type of the reformer geometries. The three type of geometries were analyzed (1D, 2D and 3D) to investigate it effect on characteristics of steam methane reforming: temperature, mole fraction and diffusion flux. The obtained results of the present numerical study were compared with experimental data and numerical results of other authors. To compare the results that are obtained for different geometries, the size of elementary cell of mesh for two-dimensional computational domain was determined. The size of edge of elementary cells is $2 \cdot 10^{-2}$ *mm*, in which a further decrease in their size did not have a significant effect on the final results. Therefore, to save computational time and use same mesh structure for all geometries the mesh structures with 1230 cells for 1D computational domain, 102,378 cells for 2D computational domain and 2,340,761 cells for 3D computational domain were chosen.

The study was performed over a wide range of residence time (from 2 $kg_{cat}s/mol_{CH4}$ to 8 $kg_{cat}s/mol_{CH4}$) for identical initial and boundary conditions for all geometries. It was found that for a reformer with a relative length (length-to-radius ratio) of 15, there are discrepancies in the mole fraction profile of the reformate products and the average temperature of the reaction mixture. With the increase in residence time, these discrepancies decrease and in the area near the reformer outlet these discrepancies are insignificant. For residence time of about 8 $kg_{cat}s/mol_{CH4}$ the discrepancies of the mole fraction of *j* element and the average temperature are practically absent.

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The discrepancy between results for 1D, 2D and 3D geometries shown in all figures above can be explained by the mass diffusion of *j* elements and thermal diffusion. Mixing and more generally diffusion phenomena, that takes place at steam methane reforming process, are never purely 2D and even less 1D. 3D diffusion is accounted for by the transport equation of turbulence quantities for the used RANS model and for laminar conditions every parameters depend on geometry of computational domain.

It was found that for a reformer that is filled with a nickel based catalyst, the differences between the results for 1D and 3D geometry become insignificant at the residence time of about 8 $kg_{cat}s/mol_{CH4}$ and the relative length of 15. The difference between the results for 2D and 3D geometry is not significant as for 1D and 3D geometries. Almost similar results are achieved with residence time 4 $kg_{cat}s/mol_{CH4}$ and relative length 15. Therefore, for engineering calculations of steam methane reforming, it is sufficient to use a one-dimensional model if the residence time is more than 8 $kg_{cat}s/mol_{CH4}$ and the relative length is more than 15. The three-dimensional and two-dimensional model should be used if the residence time and the relative length have a small value.

REFERENCES

[1] Tran A, Aguirre A, Durand H, Crose M, Christofides PD. Cfd modeling of a industrial-scale steam methane reforming furnace. Chem Eng Sci 2017;171:576–98.
[2] Nobandegani MS, Birjandi MRS, Darbandi T, Khalilipour MM, Shahraki F, Mohebbi-Kalhori D. An industrial steam methane reformer optimization using response surface methodology. J Nat Gas Sci Eng 2016;36:540–9.
[3] Diglio G, Hanak DP, Bareschino P, Mancusi E, Pepe F, Montagnaro F, et al. Techno-economic analysis of sorption-enhanced steam methane reforming in a fixed bed reactor network integrated with fuel cell. J Power Sources 2017;364:41–51.
[4] Pashchenko D. First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming. Energy 2018;143:478–87.
[5] Popov S, Svistunov I, Garyaev A, Serikov E, Temyrkanova E. The use of thermochemical recuperation in an industrial plant. Energy 2017;127:44–51.
[6] Verkhivker G, Kravchenko V. The use of chemical recuperation of heat in a power plant. Energy 2004;29(3):379–88.
[7] Yang W, Fan A, Yao H, Liu W. Effect of reduced pressures on the combustion efficiency of lean h2/air flames in a micro cavity-combustor. Int J Hydrogen Energy 2016;41(34):15354–61.
[8] Pashchenko D. Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion. Therm Eng 2013;60(6):438–43.
[9] Pashchenko D. Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation. Int J Hydrogen Energy 2017;42(22):14926–35.
[10] Nieva MA, Villaverde MM, Monzón A, Garetto TF, Marchi AJ. Steam-methane reforming at low temperature on nickel-based catalysts. Chem Eng J 2014;235:158–66.
[11] Song C, Liu Q, Ji N, Kansha Y, Tsutsumi A. Optimization of steam methane reforming coupled with pressure swing adsorption hydrogen production process by heat integration. Appl Energy 2015;154:392–401.
[12] Hafizi A, Rahimpour M, Hassanajili S. Hydrogen production via chemical looping steam methane reforming process: effect of cerium and calcium promoters on the performance of Fe2O3/Al2O3 oxygen carrier. Appl Energy 2016;165:685–94.
[13] Yoo J, Park S, Song JH, Yoo S, Song IK. Hydrogen production by steam reforming of natural gas over butyric acid-assisted nickel/alumina catalyst. Int J Hydrogen Energy 2017;42(47):28377–85.
[14] Dispenza G, Sergi F, Napoli G, Randazzo N, Di Novo S, Micari S, et al. Development of a solar powered hydrogen fueling station in smart cities applications. Int J Hydrogen Energy 2017;42(46):27884–93.
[15] Aminov R, Bairamov A. Performance evaluation of hydrogen production based on off-peak electric energy of the nuclear power plant. Int J Hydrogen Energy 2017;42(34):21617–25.
[16] Aminov R, Schastlivtsev A, Bairamov A. On the issue of investigating the kinetics of processes in dissociated water steam. Int J Hydrogen Energy 2017;42(32):20843–8.
[17] Zhang L, Tang Y, Duan Y, Hou L, Cui L, Yang F, et al. Green production of hydrogen by hydrolysis of graphene-modified aluminum through infrared light irradiation. Chem Eng J 2017;320:160–7.
[18] Soria A, Szabo L, Russ P, Suwala W, Hidalgo I, Purwanto A, World energy technology outlook 2050, European Commission, Joint Research Centre 21.
[19] Watanabe F, Kaburaki I, Shimoda N, Satokawa S. Influence of nitrogen impurity for steam methane reforming over noble metal catalysts. Fuel Process Technol 2016;152:15–21.
[20] Kumar A, Baldea M, Edgar TF. A physics-based model for industrial steam-methane reformer optimization with non-uniform temperature field. Comput Chem Eng 2017. https://doi.org/10.1016/j.compchemeng.2017.01.002.
[21] Herce C, Cortés C, Stendardo S. Computationally efficient cfd model for scale-up of bubbling fluidized bed reactors applied to sorption-enhanced steam methane reforming. Fuel Process Technol 2017;167:747–61.
[22] Abbas S, Dupont V, Mahmud T. Modelling of h2 production in a packed bed reactor via sorption enhanced steam methane reforming process. Int J Hydrogen Energy 2017;42(30):18910–21.
[23] Abbas S, Dupont V, Mahmud T. Kinetics study and modelling of steam methane reforming process over a NiO/Al2O3 catalyst in an adiabatic packed bed reactor. Int J Hydrogen Energy 2017;42(5):2889–903.
[24] Xu J, Froment GF. Methane steam reforming, methanation and water-gas shift: I. intrinsic kinetics. AIChE J 1989;35(1):88–96.
[25] Hou K, Hughes R. The kinetics of methane steam reforming over a Ni/α-Al2O catalyst. Chem Eng J 2001;82(1):311–28.
[26] Wang H, Blaylock DW, Dam AH, Liland SE, Rout KR, Zhu Y-A, et al. Steam methane reforming on a ni-based bimetallic catalyst: density functional theory and experimental studies of the catalytic consequence of surface alloying of Ni with Ag. Catal Sci Technol 2017;7(8):1713–25.
[27] Yoon Y, Kim H, Lee J. Enhanced catalytic behavior of ni alloys in steam methane reforming. J Power Sources 2017;359:450–7.
[28] Di Giuliano A, Girr J, Massacesi R, Gallucci K, Courson C. Sorption enhanced steam methane reforming by ni–cao materials supported on mayenite. Int J Hydrogen Energy 2017;42(19):13661–80.
[29] W H, El Hawa A, Lundin S-TB, Patki NS, Way JD. Steam methane reforming in a pd au membrane reactor: long-term assessment. Int J Hydrogen Energy 2016;41(24):10193–201.
[30] Bobrova LN, Sadykov VA, Mezentseva NV, Pelipenko VV, Vernikovskaya NV, Klenov OP, et al. Catalytic performance of

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structured packages coated with perovskite-based nanocomposite in the methane steam reforming reaction. Int J Hydrogen Energy 2016;41(8):4632–45.
[31] Cho EH, Koo KY, Lee HW, Park Y-K, Yoon WL, Ko CH. Preparation of egg-shell-type ni/ru bimetal alumina pellet catalysts: steam methane reforming for hydrogen production. Int J Hydrogen Energy 2017;42(29):18350–7.
[32] Chakrabarti R, Colby JL, Schmidt LD. Effects of biomass inorganics on rhodium catalysts: I. steam methane reforming. Appl Catal B Environ 2011;107(1):88–94.
[33] Kim C-H, Jae-Yun H, Hankwon L, Leec K-Y, Shin-Kun R. Methane steam reforming using a membrane reactor equipped with a pd-based composite membrane for effective hydrogen production. Int J Hydrogen Energy 2017. https://doi.org/10.1016/j.ijhydene.2017.10.054.
[34] Jaiswar VK, Katheria S, Deo G, Kunzru D. Effect of pt doping on activity and stability of ni/mgal2o4 catalyst for steam reforming of methane at ambient and high pressure condition. Int J Hydrogen Energy 2017;42(30):18968–76.
[35] Butler JJ, Nasa report to the 15th meeting of the gsics executive panel.
[36] Pulliam TH. Development of implicit methods in CFD NASA Ames research center 1970s–1980s. Comput Fluids 2011;41(1):65–71.
[37] Rzehak R, Kriebitzsch S. Multiphase cfd-simulation of bubbly pipe flow: a code comparison. Int J Multiphas Flow 2015;68:135–52.
[38] Malik MR, Revolutionary computational aerosciences (rca).
[39] Slotnick J, Khodadoust A, Alonso J, Darmofal D, Gropp W, Lurie E, et al., Cfd vision 2030 study: a path to revolutionary computational aerosciences.
[40] Mrakin A, Akimova G. Complex analysis of the efficiency of plants for the thermochemical conversion of solid fuel. Solid Fuel Chem 2015;49(4):261–5.
[41] Ku X, Li T, Løvås T. Cfd–dem simulation of biomass gasification with steam in a fluidized bed reactor. Chem Eng Sci 2015;122:270–83.
[42] Simonov V, Mrakin A, Selivanov A, Morev A, Afanaseva O. Development of an algorithm for calculating the parameters of a drum reactor for low-temperature carbonization of oil shale with ash-based heat-transfer agent. Chem Petrol Eng 2017;52(9–10):675–81.
[43] Lao L, Aguirre A, Tran A, Wu Z, Durand H, Christofides PD. Cfd modeling and control of a steam methane reforming reactor. Chem Eng Sci 2016;148:78–92.
[44] Dixon AG. Local transport and reaction rates in a fixed bed reactor tube: endothermic steam methane reforming. Chem Eng Sci 2017;168:156–77.
[45] Hoang D, Chan S, Ding O. Kinetic and modelling study of methane steam reforming over sulfide nickel catalyst on a gamma alumina support. Chem Eng J 2005;112(1):1–11.
[46] Tran A, Aguirre A, Crose M, Durand H, Christofides PD. Temperature balancing in steam methane reforming furnace via an integrated cfd/data-based optimization approach. Comput Chem Eng 2017;104:185–200.
[47] Chen J, Yan L, Song W, Xu D. Methane steam reforming thermally coupled with catalytic combustion in catalytic microreactors for hydrogen production. Int J Hydrogen Energy 2017;42(1):664–80.
[48] Corigliano O, Fragiacomo P. Numerical simulations for testing performances of an indirect internal co 2 reforming solid oxide fuel cell system fed by biogas. Fuel 2017;196:378–90.
[49] Unseld M, Szepanski C, Lindermeir A, Maus-Friedrichs W, Dahle S. Desulfurization of biogas via dielectric barrier discharges. Chem Eng Technol 2017;40(2):333–9.
[50] Salehi M-S, Askarishahi M, Godini HR, Schomacker R, Wozny G. Cfd simulation of oxidative coupling of methane in fluidized-bed reactors: a detailed analysis of flow-reaction characteristics and operating conditions. Ind Eng Chem Res 2016;55(5):1149–63.
[51] Pashchenko D. Comparative analysis of hydrogen/air combustion cfd-modeling for 3d and 2d computational domain of micro-cylindrical combustor. Int J Hydrogen Energy 2017;30(1):29545–56.
[52] Pashchenko D. Numerical investigation of high temperature synthesis gas premixed combustion via ansys fluent. In: MATEC web of conferences, vol. 145. EDP Sciences; 2018. p. 03012.
[53] Ergun S. Fluid flow through packed columns. Chem Eng Prog 1952;48:89–94.
[54] Lücke M. Influence of the Dufour effect on convection in binary gas mixtures. Phys Rev E 1995;52(1):642.
[55] Batchelor GK. An introduction to fluid dynamics. Cambridge university press; 2000.
[56] Chen L, Xu S-J, Ni M-J. The kármán vortex street inversion and heat transfer around a square cylinder at low reynolds and magnetic interaction numbers. Int J Heat Mass Tran 2018;117:768–79.
[57] Yilmaz H, Cam O, Tangoz S, Yilmaz I. Effect of different turbulence models on combustion and emission characteristics of hydrogen/air flames. Int J Hydrogen Energy 2017;42(40):25744–55.
[58] Roy RN, Sreedhara S. Modelling of methanol and H2/CO bluff-body flames using rans based turbulence models with conditional moment closure model. Appl Therm Eng 2016;93:561–70.
[59] Mrakin A, Selivanov A, Batrakov P, Sotnikov D. The chemical energy unit partial oxidation reactor operation simulation modeling. In: Journal of physics: conference series, vol. 944. IOP Publishing; 2018. p. 012082.
[60] Pashchenko D. Ansys fluent cfd modeling of solar air-heater thermoaerodynamics. Appl Sol Energy 2018;54(1):32–9.
[61] Yakhot V, Orszag SA. Renormalization group analysis of turbulence. i. basic theory. J Sci Comput 1986;1(1):3–51.
[62] Nguyen VN, Deja R, Peters R, Blum L. Methane/steam global reforming kinetics over the ni/ysz of planar pre-reformers for sofc systems. Chem Eng J 2016;292:113–22.
[63] Barth T, Jespersen D. The design and application of upwind schemes on unstructured meshes. In: 27th Aerospace sciences meeting; 1989. p. 366.

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